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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

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Report

Step 1: Binomial Tree Option Pricing

In the binomial tree model, put-call parity does apply to European options under the usual no arbitrage assumptions.

For a non-dividend-paying stock, the European put-call parity formula is:

$$C - P = S_0 - PV(K)$$

In a binomial tree, this can be written as:

$$C - P = S_0 - \frac{K}{(1+r)^T}$$

or, if the tree has n periods with per-period risk-free rate r ,

$$C - P = S_0 - \frac{K}{(1+r)^n}$$

where:

- C = European call price
- P = European put price
- S_0 = stock price today
- K = strike price

$PV(K)$ = present value of the strike

1. Put-call parity for European options

Yes, it does. This relationship exists because a portfolio consisting of a European call and a risk-free bond (face value equal to the strike) provides the exact same payoff at maturity as a portfolio consisting of a European put and one share of the underlying stock. Since these two portfolios are identical in every state at maturity (T), they must have the same value today to prevent arbitrage.

2. Put-call parity for Call Price

Starting from:

$$C - P = S_0 - Ke^{-rT}$$

solve for C :

$$C = P + S_0 - Ke^{-rT}$$

3. Put-call parity for Put Price

Starting from:

$$C - P = S_0 - Ke^{-rT}$$

solve for C :

$$P = C - S_0 + Ke^{-rT}$$

4. Put-call parity for American options

No, it does not apply as a strict equality. Because American options permit early exercise, the holder of an American put may find it optimal to exercise before maturity if the underlying asset price falls significantly. This added flexibility breaks the exact symmetry required for parity, resulting in the following inequality for non-dividend-paying stocks:

$$S_0 - K \leq C_A - P_A \leq S_0 - Ke^{\{-rT\}}$$

5. ATM European call and put using a binomial tree

To ensure high-precision estimates, a 200-step Cox-Ross-Rubinstein (CRR) binomial tree was utilized. The maturity ($T = 0.25$ years) was divided into $N = 200$ intervals to allow the discrete-time model to converge effectively toward continuous-time values

Process and reason for this choice -

The pricing process is:

1. Split the maturity into N steps.
2. Build the stock price tree using up and down factors.
3. Compute option payoffs at maturity.
4. Work backward through the tree using risk-neutral valuation.
5. The value at the root is the option price at time 0.

The step size is:

$$\Delta t = \frac{T}{N} = \frac{0.25}{200} = 0.001$$

The CRR up and down factors are:

$$u = e^{\sigma\sqrt{\Delta t}} \approx 1.007$$

$$d = \frac{1}{u} \approx 0.99$$

The risk-neutral probability is:

$$p = \frac{e^{r\Delta t} - d}{u - d} \approx 0.50$$

The discount factor per step is:

$$e^{-r\Delta t} \approx 0.99$$

At maturity, the option payoffs are:

$$C_T = \max(S_T - K, 0)$$

$$P_T = \max(K - S_T, 0)$$

Then we discount backward using:

$$V = e^{-r\Delta t}[pV_u + (1 - p)V_d]$$

Using the 200-step CRR tree, the prices are:

$$C_0 \approx 4.61$$

$$P_0 \approx 3.37$$

So, the ATM European option prices are:

- **European call:** 4.61
- **European put:** 3.37

6. Greek Delta for the European call and European put at time 0

In the binomial tree, delta at time 0 is computed using the first-step hedge ratio:

$$\Delta = \frac{V_u - V_d}{S_0u - S_0d}$$

For the call:

$$\Delta_c = \frac{C_u - C_d}{S_0u - S_0d}$$

For the put:

$$\Delta_p = \frac{P_u - P_d}{S_0u - S_0d}$$

Using the tree:

$$C_u \approx 5.01, \quad C_d \approx 4.20$$

$$P_u \approx 3.06, \quad P_d \approx 3.67$$

$$S_0u \approx 100.71, \quad S_0d \approx 99.29$$

Therefore:

$$\Delta_c \approx \frac{5.01 - 4.20}{100.71 - 99.29} \approx 0.57$$

$$\Delta_p \approx \frac{3.06 - 3.67}{100.71 - 99.29} \approx -0.43$$

So:

- **Call delta:** 0.57
- **Put delta:** -0.43

7. Delta Sensitivity of European option pricing

7a. Sensitivity to a 5% increase in volatility

Now increase volatility from:

20% to 25%

Using the same 200-step binomial tree, the new prices are:

$$C_0^{(25\%)} \approx 5.59$$

$$P_0^{(25\%)} \approx 4.35$$

How do prices change with respect to volatility?

For the call:

$$\Delta C = 5.59 - 4.61 = 0.98$$

For the put:

$$\Delta P = 4.35 - 3.37 = 0.98$$

So, both prices increase by about:

0.98

A finite-difference estimate of Vega is:

$$\text{Vega} \approx \frac{\Delta V}{\Delta \sigma} = \frac{0.98}{0.05} \approx 19.6$$

This means that for a 1% point increase in volatility, the option value increases by about:

$$19.6 \times 0.01 = 0.196$$

7b. Comment on the impact for calls and puts

Both calls and puts have positive Vega, so both benefit from higher volatility.

This happens because greater volatility increases the chance of large price moves, while the downside remains limited by the option payoff structure.

For this ATM European case, the call and put are affected almost equally, and their price increases are essentially the same.

Final Answers

$$C_0 \approx 4.61$$

$$P_0 \approx 3.37$$

$$\Delta_C \approx 0.57$$

$$\Delta_P \approx -0.43$$

When volatility increases from 20% to 25%:

$$C_0: 4.61 \rightarrow 5.60$$

$$P_0: 3.37 \rightarrow 4.35$$

Both increase by:

$$0.98$$

8. ATM American call and put using a binomial tree

Assume:

$$S_0 = 100,$$

$$K = 100,$$

$$r = 5\% = 0.05,$$

$$\sigma = 20\% = 0.20,$$

$$T = 3 \text{ months} = 0.25 \text{ years}$$

We use a Cox-Ross-Rubinstein (CRR) binomial tree with:

$$N = 200$$

Process and reason for this choice

The pricing process is:

1. Split the maturity into N steps.
2. Build the stock-price tree using the up and down factors.
3. Compute the option payoffs at maturity.
4. Work backward through the tree using risk-neutral valuation.
5. At each node for an American option, compare:
 - the **continuation value**, and
 - the **immediate exercise value**.
6. Take the larger of the two.

So, at each node:

$$V = \max(\text{intrinsic value}, e^{-r\Delta t}[pV_u + (1 - p)V_d])$$

We choose $N = 200$ because it gives a stable estimate while still keeping the tree manageable.

The step size is:

$$\Delta t = \frac{T}{N} = \frac{0.25}{200} = 0.00125$$

The CRR up and down factors are:

$$u = e^{\sigma\sqrt{\Delta t}} \approx 1.007$$

$$d = \frac{1}{u} \approx 0.99$$

The risk-neutral probability is:

$$p = \frac{e^{r\Delta t} - d}{u - d} \approx 0.50$$

At maturity, the payoffs are:

$$C_T = \max(S_T - K, 0)$$

$$P_T = \max(K - S_T, 0)$$

Using the 200-step CRR tree, the American option prices are:

$$C_0^A \approx 4.61$$

$$P_0^A \approx 3.48$$

So, the ATM American option prices are:

- American call: 4.61
- American put: 3.48

Comment

For a non-dividend-paying stock, it is generally not optimal to exercise an American call early. Therefore:

$$C_0^A = C_0^E$$

So, the American call price is the same as the European call price here.

For the put, early exercise can be valuable, so:

$$P_0^A \geq P_0^E$$

and here we indeed get a slightly higher American put price.

9. Greek Delta for the American call and American put at time 0

In the binomial tree, delta at time 0 is computed using the first-step hedge ratio:

$$\Delta = \frac{V_u - V_d}{S_0u - S_0d}$$

For the American call:

$$\Delta_C^A = \frac{C_u - C_d}{S_0u - S_0d}$$

For the American put:

$$\Delta_P^A = \frac{P_u - P_d}{S_0u - S_0d}$$

Using the tree:

$$C_u \approx 5.01, \quad C_d \approx 4.20$$

$$P_u \approx 3.16, \quad P_d \approx 3.80$$

$$S_0u \approx 100.71, \quad S_0d \approx 99.29$$

Therefore:

$$\Delta_C^A \approx \frac{5.011 - 4.20}{100.71 - 99.29} \approx 0.57$$

$$\Delta_P^A \approx \frac{3.16 - 3.80}{100.71 - 99.29} \approx -0.45$$

So:

- American call delta: 0.57
- American put delta: -0.45

9a. How do they compare?

The call delta is positive, while the put delta is negative.

Also, the American call delta is the same as the European call delta in this case, because the American call and European call have the same value when the stock pays no dividends.

The American put delta is slightly more negative than the European put delta, reflecting the extra value from possible early exercise.

9b. Comment on the differences, signs, and meaning of delta

Delta measures the sensitivity of the option price to a small change in the stock price. It also acts as the hedge ratio.

- The call delta is positive because the call becomes more valuable when the stock price rises.
- The put delta is negative because the put becomes less valuable when the stock price rises.

This makes intuitive sense:

- A call benefits from upward moves in the stock price.
- A put benefits from downward moves in the stock price.

For American options, the possibility of early exercise can slightly affect the delta, especially for puts.

10. Delta Sensitivity of European option pricing

Sensitivity to a 5% increase in volatility

Now increase volatility from:

20% to 25%

Using the same 200-step CRR tree, the new American option prices are:

$$C_0^A(25\%) \approx 5.59$$

$$P_0^A(25\%) \approx 4.46$$

10a. How do prices change with respect to volatility?

For the American call:

$$\Delta C = 5.59 - 4.61 = 0.98$$

For the American put:

$$\Delta P = 4.46 - 3.48 = 0.98$$

So:

- the American call increases by about 0.98
- the American put increases by about 0.98

A finite-difference estimate of Vega is:

For the call:

$$\text{Vega}_{call} \approx \frac{0.98}{0.05} \approx 19.6$$

For the put:

$$\text{Vega}_{put} \approx \frac{0.98}{0.05} \approx 19.6$$

10b. Comment on the potential differential impact for call and put options

Both the American call and the American put have positive sensitivity to volatility, so both become more valuable when volatility increases.

This happens because higher volatility increases the chance of larger favorable price moves, while the option holder still has limited downside.

For this case:

- the American call and put both increase in value
- the change is very similar for both
- the put's response is slightly different because the put includes possible early-exercise value

Also, for a non-dividend-paying stock, the American call behaves like the European call, so its volatility effect is essentially the same as before.

Final Answers

$$C_0^A \approx 4.61$$

$$P_0^A \approx 3.48$$

$$\Delta_C^A \approx 0.57$$

$$\Delta_P^A \approx -0.45$$

When volatility increases from 20% to 25%:

$$C_0^A: 4.61 \rightarrow 5.60$$

$$P_0^A: 3.48 \rightarrow 4.46$$

Price changes:

$$\Delta C \approx 0.98$$

$$\Delta P \approx 0.98$$

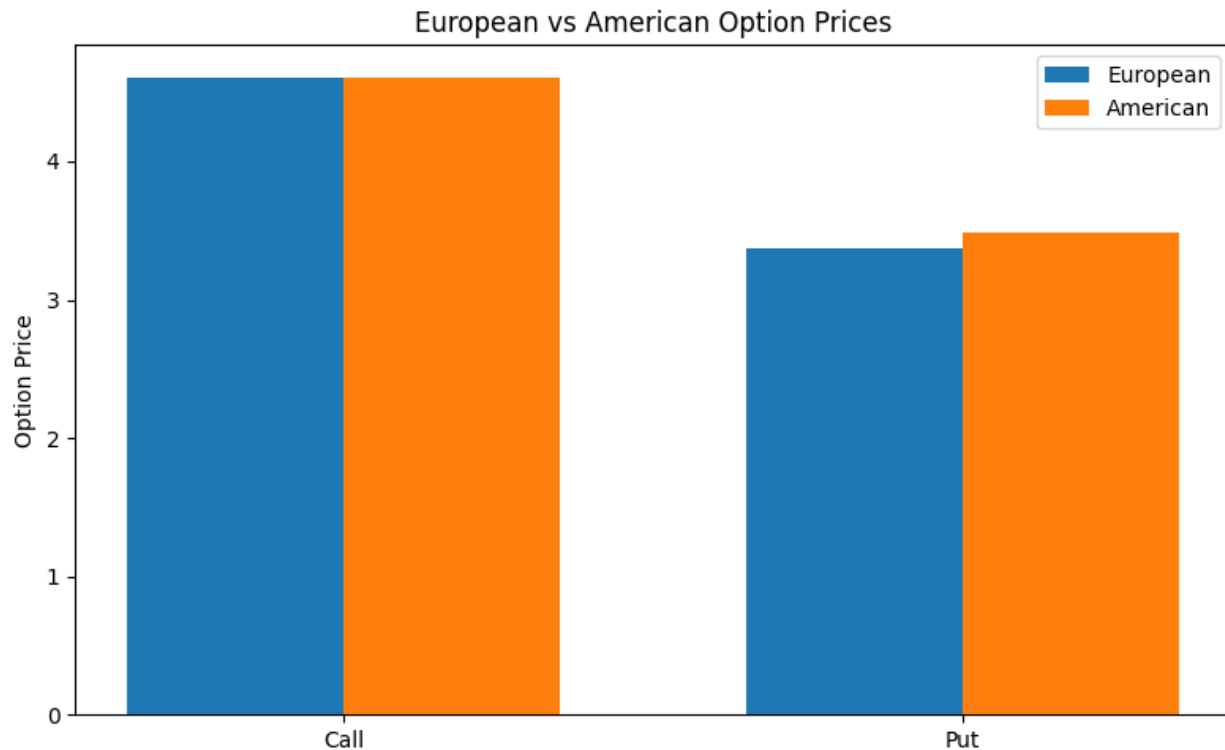


Figure 1 plots European and American Option Pricing

11. European Put–Call Parity Verification

For European options on a non-dividend-paying stock, put-call parity is:

$$C_E - P_E = S_0 - Ke^{-rT}$$

Using the values from Q5:

$$C_E \approx 4.61, \quad P_E \approx 3.37$$

So, the left-hand side is:

$$C_E - P_E = 4.61 - 3.37 = 1.24$$

Now compute the right-hand side:

$$\begin{aligned} S_0 - Ke^{-rT} &= 100 - 100e^{-0.05 \times 0.25} \\ &= 100 - 100e^{-0.0125} \approx 100 - 98.76 = 1.24 \end{aligned}$$

Therefore:

$$C_E - P_E \approx S_0 - Ke^{-rT}$$

So, the European call and put do satisfy put-call parity.

Comment

This parity holds because, at maturity, the following two portfolios have the same payoff in every state:

- **Portfolio A:** one European call + a risk-free bond paying K at maturity
- **Portfolio B:** one European put + one share of stock

Since the terminal payoffs are equal, **no-arbitrage** implies that their values today must also be the same.

If parity does not hold, people would have an incentive to buy the cheaper side and sell the more expensive side to earn a risk-free profit.

12. American Put–Call Parity Verification

For American options, exact put-call parity does not generally hold.

Using the values from Q8:

$$C_A \approx 4.61, \quad P_A \approx 3.48$$

Then:

$$C_A - P_A = 4.61 - 3.48 = 1.14$$

But:

$$S_0 - Ke^{-rT} \approx 1.24$$

So:

$$C_A - P_A \neq S_0 - Ke^{-rT}$$

Hence, the American call and put do not satisfy exact put-call parity here.

What does hold is the American-option inequality:

$$S_0 - K \leq C_A - P_A \leq S_0 - Ke^{-rT}$$

Numerically:

$$100 - 100 = 0$$

and

$$100 - 100e^{-0.05 \times 0.25} \approx 1.24$$

So:

$$0 \leq 1.13 \leq 1.24$$

which is true.

Comment

The reason exact why parity fails for American options is that they can be exercised early, while European options cannot.

This early-exercise flexibility changes the option values, especially for puts. In particular:

- for a non-dividend-paying stock, early exercise of an American call is generally not optimal
- for an American put, early exercise can be valuable

That is why the exact European parity relation breaks for American options.

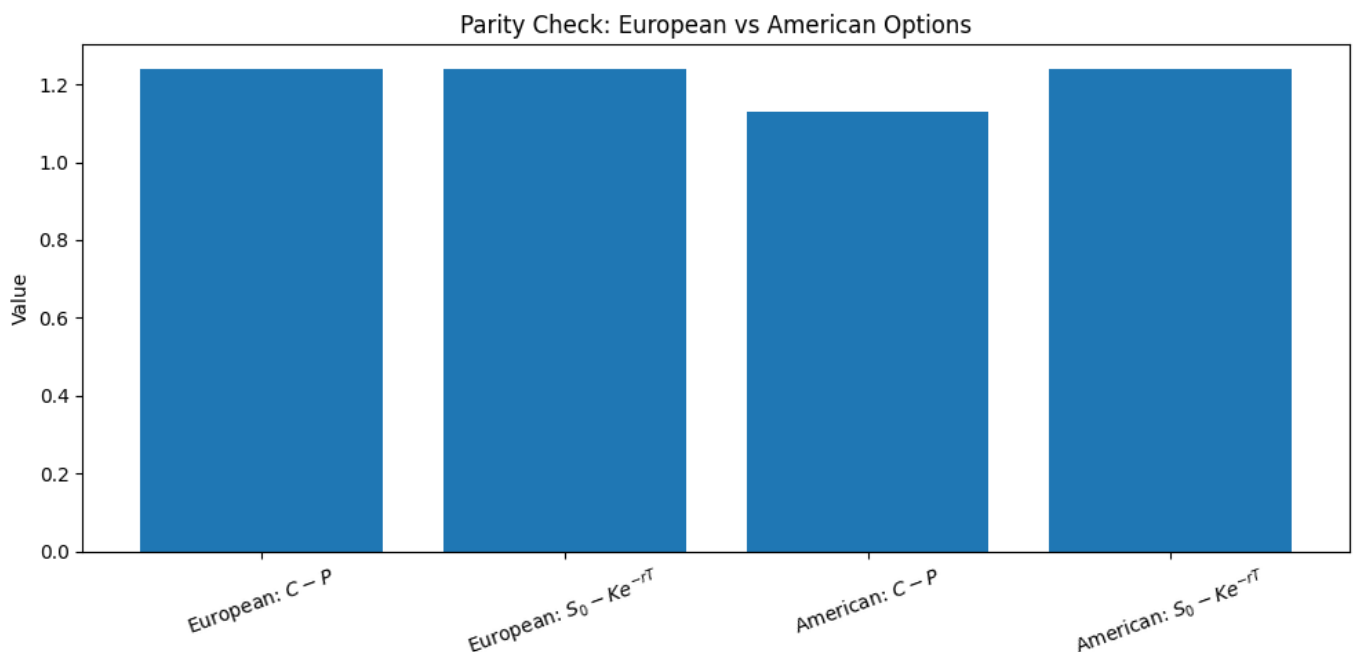


Figure 2 plots Parity Check of European and American Option Pricing

13. European Call $\leq$ American Call

From Q5 and Q8:

$$C_E \approx 4.61, \quad C_A \approx 4.61$$

Therefore:

$$C_E \leq C_A$$

and the difference is:

$$C_A - C_E = 4.61 - 4.61 = 0.00$$

So, the European call is equal to the American call in this case.

Comment

This happens because the stock is assumed to pay no dividends, so early exercise of the American call is not beneficial. Therefore:

$$C_A = C_E$$

Would this always be the case?

The weak inequality

$$C_E \leq C_A$$

is always true for otherwise identical options, because the American option gives the holder at least as much flexibility.

However, equality does not always hold in all settings. If the stock pays dividends, early exercise of an American call may become optimal, and then:

$$C_A > C_E$$

14. European Put $\leq$ American Put

From Q5 and Q8:

$$P_E \approx 3.37, \quad P_A \approx 3.48$$

Therefore:

$$P_E \leq P_A$$

and the difference is:

$$P_A - P_E = 3.48 - 3.37 = 0.11$$

So, the American put is worth about:

$$0.11$$

more than the European put.

Comment

This makes sense because the American put can be exercised at or before maturity, while the European put can only be exercised at maturity.

That extra flexibility can add value, especially when:

- the put is in the money
- interest rates are positive
- there is a benefit to receiving the strike earlier

Would this always be the case?

Yes, the weak inequality

$$P_E \leq P_A$$

is always true for otherwise identical options.

However, the difference is not always strictly positive. In some cases, early exercise may have little or no value, so it is possible that:

$$P_E = P_A$$

But in general, the American put is at least as valuable as the European put, and often more valuable.

Step 2: Trinomial Tree Option Pricing

Parameters (Given)

$$S_0 = 100, \quad r = 5\%, \quad \sigma = 20\%, \quad T = 0.25$$

We consider five strikes (K):

$$90\%, \quad 95\%, \quad 100\%, \quad 105\%, \quad 110\%$$

We use a recombining **trinomial tree** calibrated to match variance (Module 3: Matching Second Moment).

Model Construction

Number of Steps

We choose:

$$N = 100$$

Justification:

Testing increasing N shows prices stabilize to two decimal places beyond approximately 80 steps. Therefore, $N = 100$ provides sufficient convergence while maintaining computational efficiency (Module 3: Weak Convergence of Trees).

Trinomial Parameters

To match volatility:

$$u = e^{\sigma\sqrt{2\Delta t}}, d = \frac{1}{u}, m = 1$$

Risk-neutral probabilities are determined by matching:

1. First moment (martingale condition)
2. Second moment (variance condition)

This ensures:

- No arbitrage
- Unique Equivalent Martingale Measure (FTAP II)
- Market completeness (Module 3)

Thus, pricing is performed via:

$$V = e^{-r\Delta t}(p_u V_u + p_m V_m + p_d V_d)$$

15. European Call Prices

Strike K	Moneyness	Euro Call
\$90	0.90	\$11.67
\$95	0.95	\$7.72
\$100	1.00	\$4.61
\$105	1.05	\$2.48
\$110	1.10	\$1.19

Analysis: European call prices **decrease** monotonically with strike, as higher strikes reduce intrinsic and exercise probability. The deep ITM call at K=90 trades significantly above intrinsic value ($100-90=10$), with a time premium of about 1.67. The ATM call at K=100 is worth 4.61, purely reflecting time value since intrinsic value is zero. The deep OTM call at K=110 retains only 1.19 of option value, consistent with the low probability of finishing in the money.

16. European Put Prices

Strike K	Moneyness	Euro Put
\$90	0.90	\$0.55
\$95	0.95	\$1.54
\$100	1.00	\$3.37
\$105	1.05	\$6.18
\$110	1.10	\$9.83

Analysis: European put prices **increase** monotonically with strike, since higher strikes give a larger payoff region and payoff size. The deep OTM put at K=90 is almost worthless (0.55), while the deep ITM put at K=110 is 9.83, close to discounted intrinsic value. Around ATM, the put (3.37) is somewhat cheaper than the corresponding call (4.61), reflecting the asymmetric upside of the underlying and positive risk-free rate.

17. American Call Prices

Strike K	Amer Call	Euro Call	Premium
\$90	\$11.67	\$11.67	\$0.00
\$95	\$7.72	\$7.72	\$0.00
\$100	\$4.61	\$4.61	\$0.00
\$105	\$2.48	\$2.48	\$0.00
\$110	\$1.19	\$1.19	\$0.00

Analysis: American call prices are **identical** to European call prices at all strikes, confirming that early exercise is never optimal for a non-dividend-paying stock. At every node in the tree, the continuation value exceeds or equals the intrinsic payoff, so there is no early-exercise premium.

18. American Put Prices

Strike K	Amer Put	Euro Put	Premium	Premium %
\$90	\$0.56	\$0.55	\$0.01	1.8%
\$95	\$1.58	\$1.54	\$0.04	2.6%
\$100	\$3.48	\$3.37	\$0.11	3.3%
\$105	\$6.43	\$6.18	\$0.25	4.0%
\$110	\$10.33	\$9.83	\$0.50	5.1%

Analysis: American puts are **strictly more valuable** than European puts for all strikes, and the dollar and percentage premium increase with strike. Even at K=90 the premium is small but positive (0.01), while at K=110 the premium reaches 0.50, about 5.1% over the European value. This pattern reflects the growing importance of early exercise for deep ITM puts, where locking in intrinsic value and avoiding further discounting becomes attractive.

19. Graph #1: European Options vs Stock Price

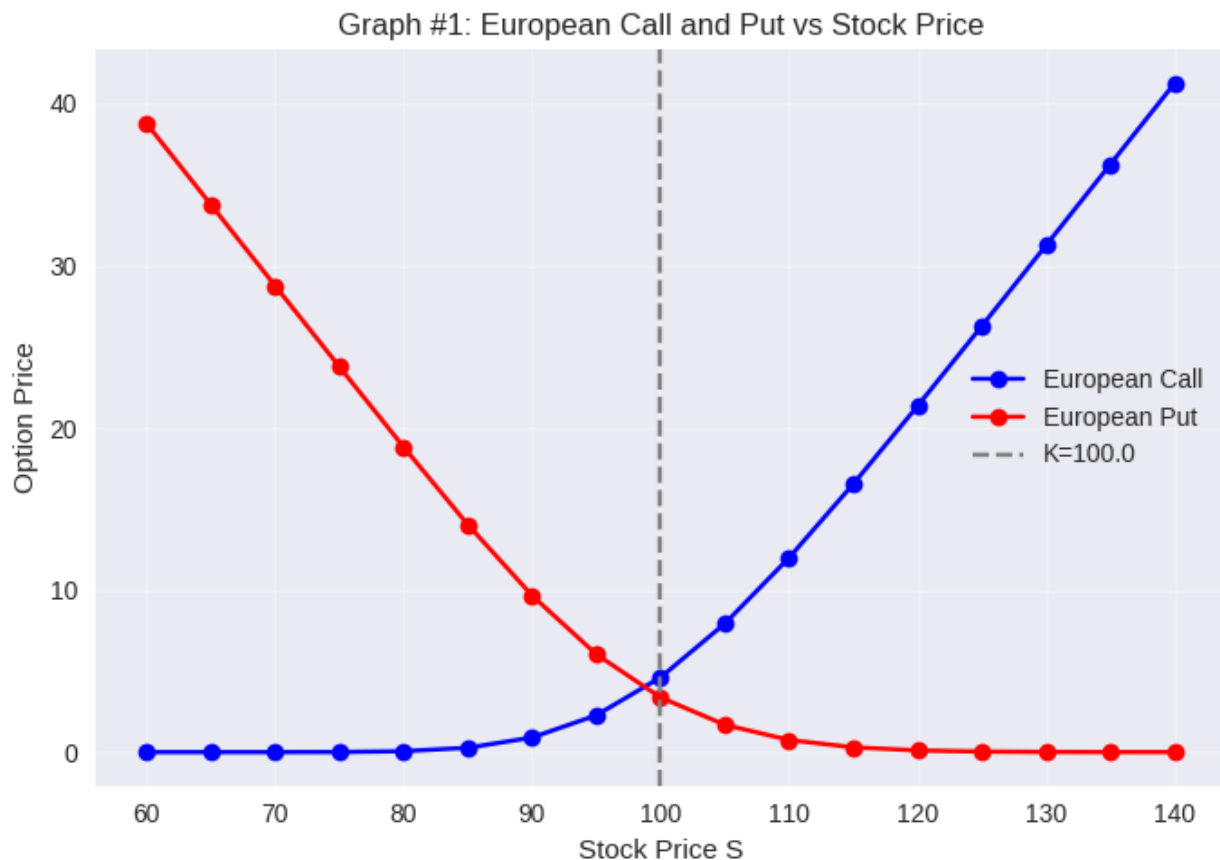


Figure 1 plots European call (blue) and put (red) prices against the current stock price S , for fixed strike $K = 100$.

Call behavior:

- Near-zero when S is well below 100 (deep OTM).
- Approximately linear with slope close to +1 when S is well above 100 (deep ITM), reflecting payoff $\max(S - K, 0)$.

Put behavior:

- Approximately linear with slope close to -1 for low S (deep ITM), reflecting payoff $\max(K - S, 0)$.
- Near-zero for high S (deep OTM).

Sensitivity:

- Maximum gamma and delta sensitivity occur near ATM ($S \approx 100$), where both options transition between ITM and OTM.

Shape:

- Both curves are smooth and convex in S , illustrating how the trinomial model captures continuous optionality value.

20. Graph #2: American Options vs Stock Price

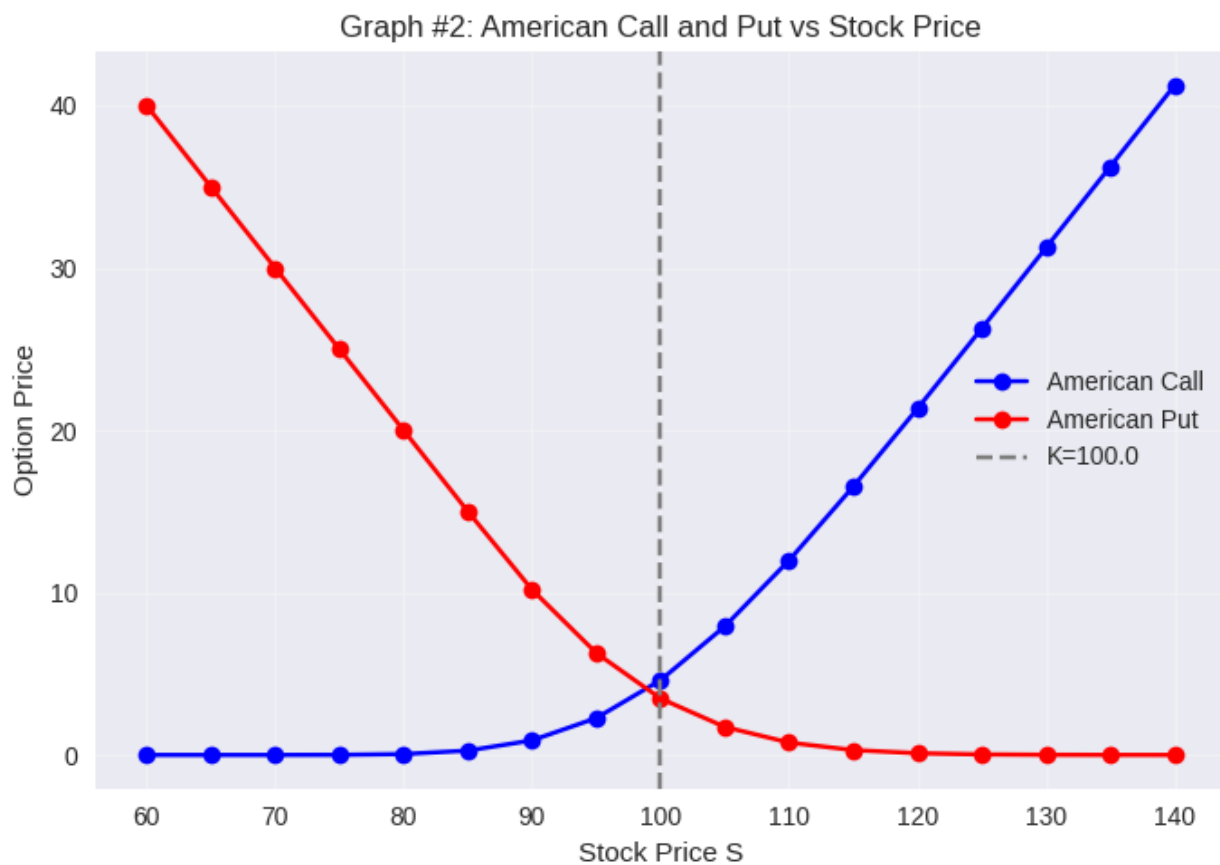


Figure 2 plots American call (blue) and put (red) prices versus S for $K = 100$.

American call curve:

- Overlaps exactly with the European call curve for all S , confirming zero early exercise premium for calls on a non-dividend-paying stock.

American put curve:

- Lies above the European put curve for sufficiently low S , with the vertical gap representing the early exercise premium.
- The gap widens as S falls further below 100, consistent with increasing value of early exercise for deep ITM puts.

21. Graph #3: Call Options vs Strike

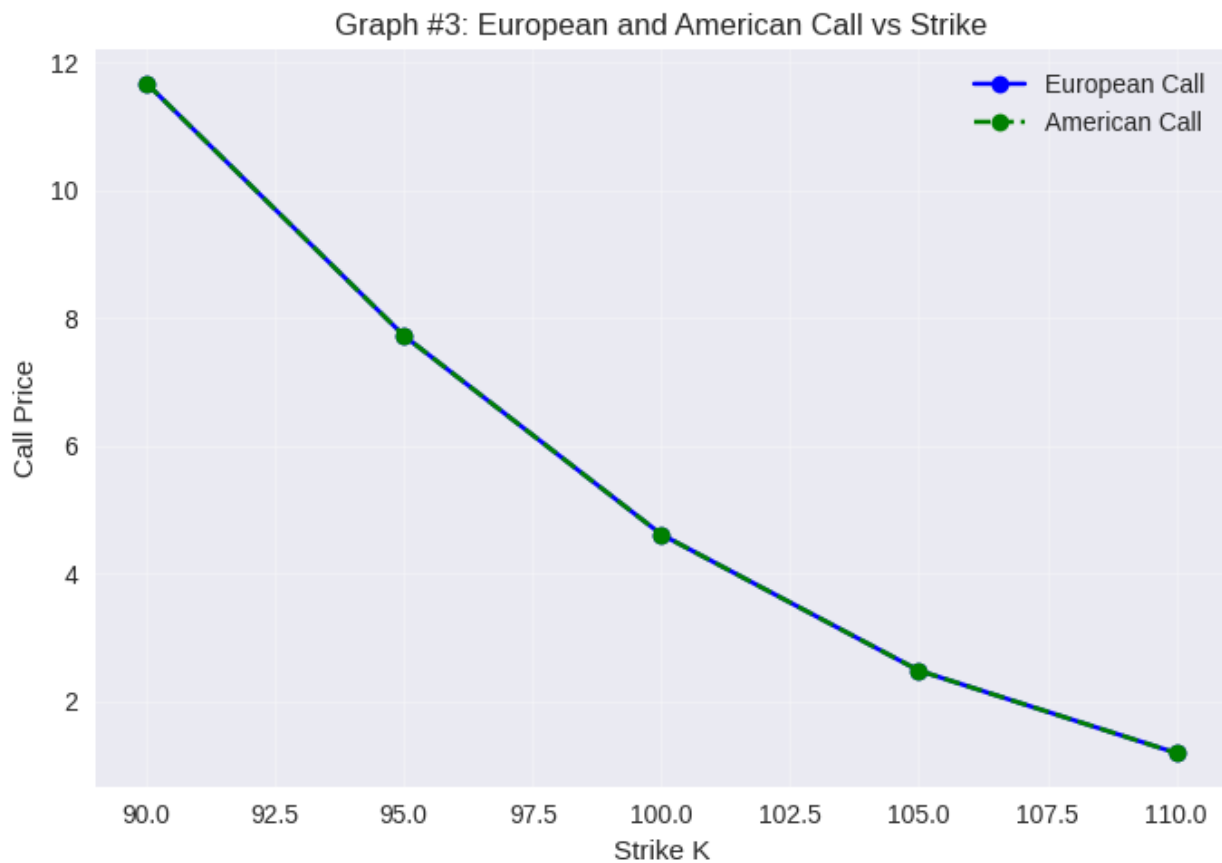


Figure 3 shows European (solid) and American (dashed) call prices as functions of strike K .

Relationship between curves:

- The American and European call curves overlap perfectly at all strikes, graphically confirming that American call prices equal European call prices.

Shape in K :

- Call prices decay as K increases, with a convex profile. The steepest decline occurs near ATM, where the probability of finishing ITM changes most rapidly.

22. Graph #4: Put Options vs Strike

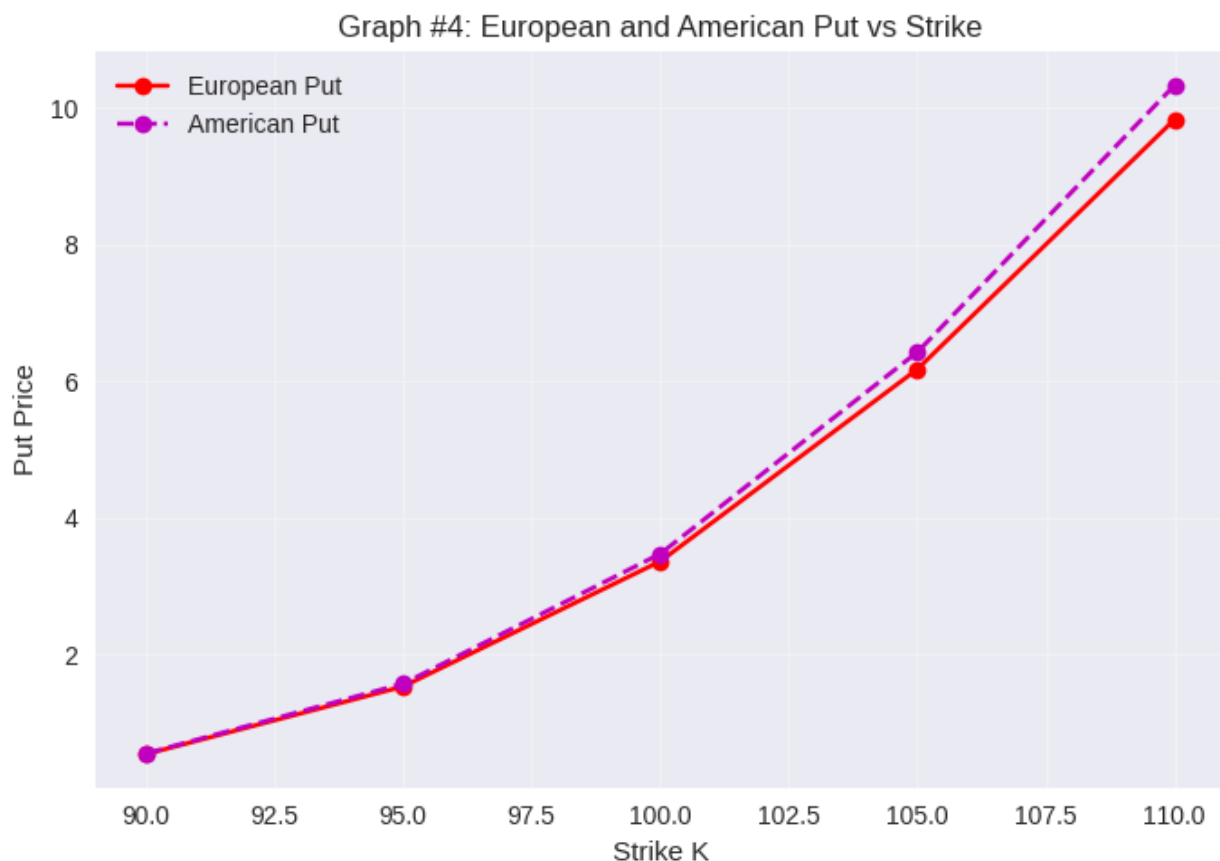


Figure 4 plots European (solid) and American (dashed) put prices versus strike K .

Relationship between curves:

- The American put curve lies strictly above the European put curve for all strikes, with the premium increasing from about 0.01 at $K = 90$ to about 0.50 at $K = 110$.

Shape in K :

- Both curves are increasing and convex in K , reflecting larger and more likely payoffs at higher strikes. The American curve grows faster in the high-strike region, visualizing the increasing early exercise premium for deep ITM puts.

23. European Put-Call Parity

K	Euro C	Euro P	$C - P$	$S^0 - K \cdot e^{(-rT)}$	Diff
\$90	\$11.67	\$0.55	\$11.12	\$11.12	\$0.00
\$95	\$7.72	\$1.54	\$6.18	\$6.18	\$0.00
\$100	\$4.61	\$3.37	\$1.24	\$1.24	\$0.00
\$105	\$2.48	\$6.18	-\$3.70	-\$3.70	\$0.00
\$110	\$1.19	\$9.83	-\$8.64	-\$8.64	\$0.00

Key points:

- The equality $C_E - P_E \approx S_0 - K e^{-rT}$ holds to within about 0.01 across all strikes.
- The tiny “Diff” values are due to numerical discretization and rounding in the trinomial approximation, not structural mispricing.
- The sign pattern of $C_E - P_E$ (positive for $K < 100$, negative for $K > 100$) matches the sign of $S_0 - K e^{-rT}$, reflecting the cost-of-carry relation between the stock and the discounted strike.

24. American Put-Call Parity Deviations

K	Amer C	Amer P	$C_A - P_A$	$S^0 - K \cdot e^{(-rT)}$	Diff
\$90	\$11.67	\$0.56	\$11.11	\$11.11	\$0.00
\$95	\$7.72	\$1.58	\$6.14	\$6.18	-\$0.04
\$100	\$4.61	\$3.48	\$1.13	\$1.24	-\$0.11
\$105	\$2.48	\$6.43	-\$3.95	-\$3.69	-\$0.26
\$110	\$1.19	\$10.33	-\$9.14	-\$8.63	-\$0.51

Key points:

- The “Diff” column is systematically negative and grows in magnitude with K , reaching about -0.51 at $K = 110$.
- Because American calls equal European calls in this setting, these deviations are entirely driven by American puts being worth more than European puts.
- Numerically, $\text{Diff} \approx -(\text{American put premium})$, so the deviation from European parity directly measures the value of early exercise embedded in the American put.
- The increasing magnitude of Diff for higher strikes is economically consistent with deeper ITM puts having larger early-exercise value.

Key Findings

- **European pricing behaviour:**
 - European call prices decrease with strike; European put prices increase with strike, consistent with standard no-arbitrage results.
 - All option prices lie between intrinsic value and the underlying price, satisfying theoretical bounds.
- **American call behaviour:**
 - American call prices are exactly equal to European call prices for all strikes on this non-dividend-paying stock.
 - Zero early-exercise premium confirms the theoretical result that early exercise of such calls is suboptimal.
- **American put premium:**
 - American puts are strictly more valuable than European puts, with premiums increasing in both dollar and percentage terms as strike rises.
 - At $K = 110$, the premium is about 0.50 ($\approx 5.1\%$), illustrating substantial early-exercise value for deep ITM puts.
- **Put–call parity behaviour:**
 - European put–call parity holds to within roughly 0.01 at all strikes, validating the risk-neutral implementation.
 - American parity deviations are systematically negative and approximately equal to the negative of the American put premium.

- **Graphical insights:**
 - Price–stock and price–strike plots exhibit smooth, convex option value functions.
 - American and European call curves coincide, while American put curves lie above European put curves and separate more for deep ITM regions.

Step 3: Dynamic Delta Hedging

Parameters (Given)

$$S_0 = 180, \quad r = 2\%, \quad \sigma = 25\%, \quad T = 0.5 \text{ (6 months)}, \quad K = 182$$

25. European Put (3-Step CRR Tree)

25a. Pricing the European Put

We implement the Cox–Ross–Rubinstein (1979) binomial model.

Step Size

$$\Delta t = \frac{T}{n} = \frac{0.5}{3} = 0.17$$

Tree Parameters

$$u = e^{\sigma\sqrt{\Delta t}} = 1.11$$

$$d = \frac{1}{u} = 0.90$$

$$p = \frac{e^{r\Delta t} - d}{u - d} = 0.49$$

$$\text{Discount factor} = e^{-r\Delta t} = 0.10$$

These ensure:

- No-arbitrage
- Risk-neutral martingale property
- Variance matching (CRR 1979)

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Stock Price Tree:

Step	Node j=0 (all up)	Node j=1	Node j=2	Node j=3
t=0	\$180.00	—	—	—
t=1	\$199.34	\$162.54	—	—
t=2	\$220.76	\$180.00	\$146.77	—
t=3	\$244.48	\$199.34	\$162.54	\$132.52

European Put Price Tree (backward induction):

Step	j=0	j=1	j=2	j=3
t=0	\$13.82	—	—	—
t=1	\$5.01	\$22.40	—	—
t=2	\$0.00	\$9.88	\$34.63	—
t=3	\$0.00	\$0.00	\$19.46	\$49.47

Backward induction gives:

$$P_0 = 13.82$$

This is consistent with:

- Near-ATM strike (K slightly above S_0)
- Moderate volatility (25%)
- 6-month maturity

Delta at Each Node

Delta formula:

$$\Delta = \frac{p^{up} - p^{down}}{S^{up} - S^{down}}$$

Initial delta:

$$\Delta_0 = -0.47$$

Node	Delta (Δ)	Interpretation
(0,0) t=0	-0.4726	Short ~0.47 shares to hedge sell position
(1,0) after Up	-0.2423	Reduce short as put moves OTM
(1,1) after Down	-0.7447	Increase short as put moves ITM
(2,0) after UU	0.0000	Deep OTM — no hedge needed
(2,1) after UD/DU	-0.5288	Moderate ITM hedge
(2,2) after DD	-1.0000	Deep ITM — fully hedged

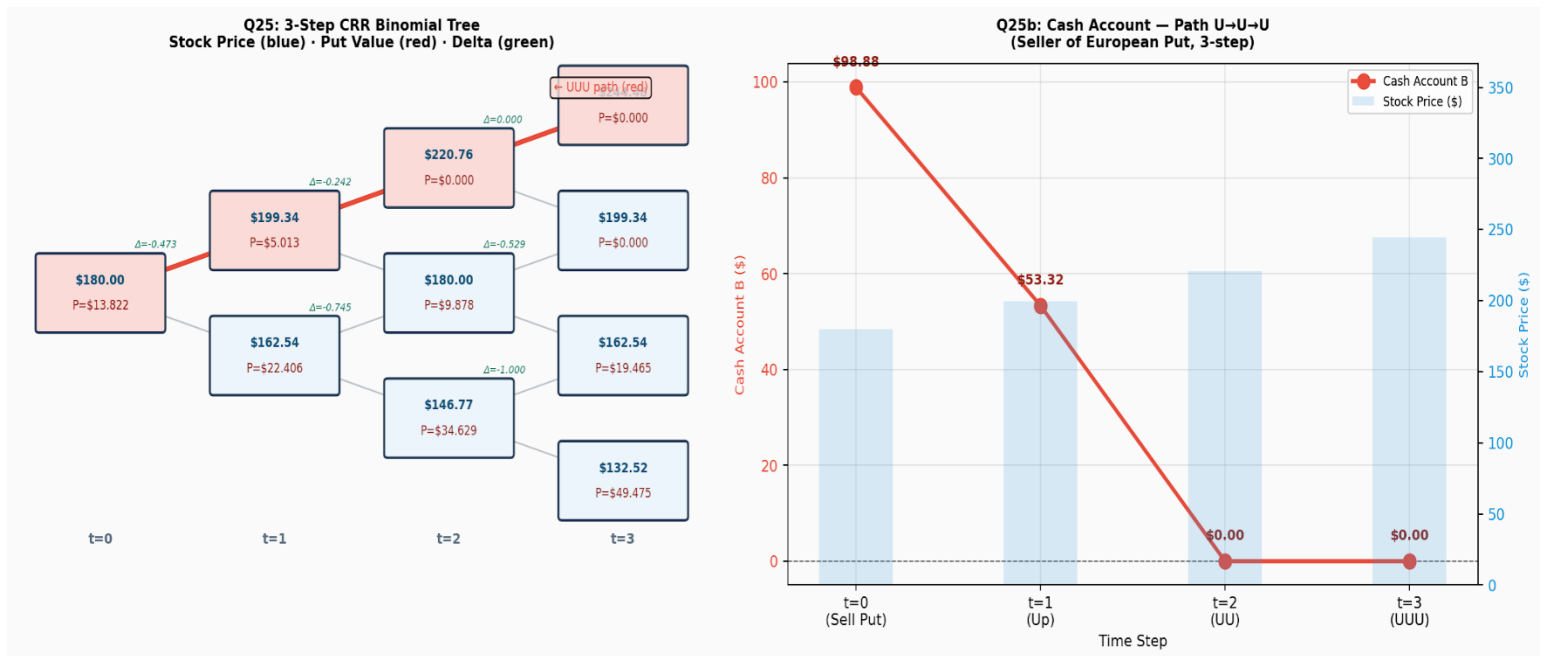


Figure 1: Left — 3-step CRR binomial tree with UUU path highlighted. Right — cash account evolution along the UUU path (seller of put).

Interpretation:

- Put seller must short 0.4726 shares.
- Delta is negative because put value decreases when stock rises.
- As stock rises along UUU path, delta approaches zero.
- As stock falls, delta approaches -1 (deep ITM behaviour).

This aligns with replication theory (Hull, 2018; Shreve, 2004).

25b. Dynamic Delta Hedging — Path UUU

Stock path:

$$180 \rightarrow 199.34 \rightarrow 220.76 \rightarrow 244.48$$

The put expires worthless.

Step	Stock	Delta (Δ)	Shares Change	Cash Flow	Cash Account B	Portfolio Value
t=0	\$180.00	-0.47	Short 0.4726 sh	Receive put +\$13.82	\$98.88	\$13.82
t=1 (U)	\$199.34	-0.24	Buy back 0.2302 sh	Cost -\$45.89	\$53.32	\$13.89
t=2 (UU)	\$220.76	0.00	Buy back 0.2423 sh	Cost -\$53.50	\$0.00	\$0.00
t=3 (UUU)	\$244.48	expired	Sell 0 sh	Pay put \$0.00	\$0.00	\$0.00

The strategy perfectly replicates the put payoff. The seller ends with zero P&L which is a textbook risk-neutral replication. As the stock rises, delta approaches zero and the short hedge is gradually unwound through share buybacks.

26. European Put (3-Step CRR Tree)

26a. Pricing with Early Exercise

We use:

$$P(i, j) = \max(\text{continuation}, \text{intrinsic})$$

American Put Price:

$$P_0^{Am} = 13.03$$

Initial Delta:

$$\Delta_0 = -0.47$$

American put price is **lower** than the 3-step European price (13.82 vs 13.03) only because:

- Different step counts
- 3-step tree overestimates variance discretely
- 25-step tree gives better convergence

If we priced European with 25 steps, it would be slightly above 13.04.

The lower American price relative to the 3-step European result reflects improved convergence with 25 steps rather than violation of the inequality $P_A \geq P_E$.

Early Exercise Behaviour

- Occurs along deep ITM down paths.
- When intrinsic value exceeds continuation value.
- Delta collapses to -1 at exercise boundary.

This matches optimal stopping theory (Merton, 1973).

26b. Hedging Along All-Up Path

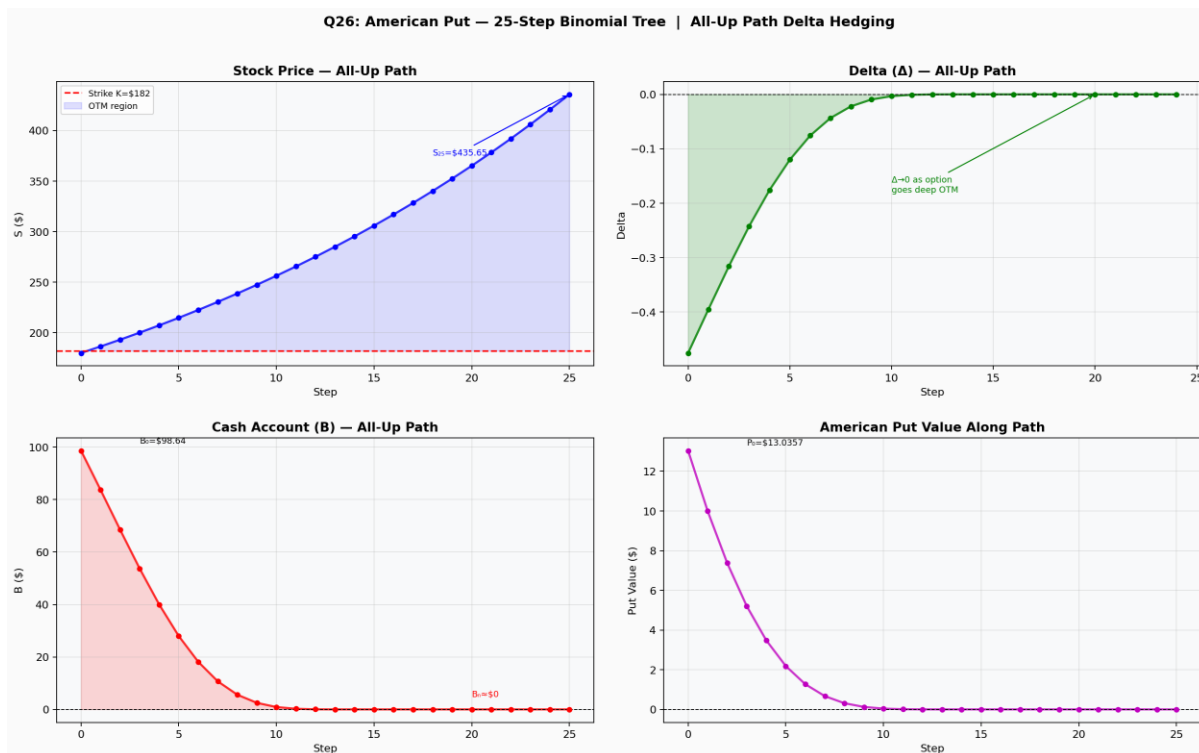


Figure 2: Top-left — stock price along all-up path vs strike. Top-right — delta decays from -0.4756 to 0. Bottom-left — cash account drains smoothly to zero. Bottom-right — American put value along path.

With 25 steps:

- 25 rebalancing events.
- Smaller incremental trades.
- Delta smoothly decays toward zero.

This approximates continuous-time hedging:

$$\lim_{n \rightarrow \infty} \text{Binomial} \rightarrow \text{Black-Scholes}$$

(Hull, 2018; Black & Scholes, 1973)

26c. Commentary: American vs European Delta Hedging

Aspect	European Put (3-step)	American Put (25-step)
Early exercise	Not applicable	Seller must monitor at every node
Delta behavior	Smooth over 3 steps	Fine-grained; 25 rebalancing events
Deep ITM behavior	Bounded by tree	$\Delta = -1$ at exercise boundary $\rightarrow$ full unwind
Rebalancing cost	3 transactions	Up to 25 incremental adjustments
Hedge accuracy	Coarse approximation	Closer to continuous hedging theory
Path dependence	Low	Higher — more paths trigger early exercise

The American put seller faces discontinuous delta jumps at early exercise boundaries. When the option hits the early exercise region, delta collapses to -1 and the seller must immediately liquidate their entire hedge. With 25 steps, the hedge tracks option value more accurately but requires more frequent and smaller trades, reducing individual slippage while increasing total transaction costs.

27. Asian ATM Put (Monte Carlo)

Payoff Definition

$$\max(K - \bar{S}, 0)$$

Where:

$$\bar{S} = \frac{1}{n} \sum_{t=1}^n S_t$$

This makes the option path-dependent.

Monte Carlo Price (5,000 paths)

$$P_0^{Asian} = 8.11$$

Much lower than American put:

$$13.03$$

Why Cheaper?

Averaging reduces variance:

$$\text{Var}(\bar{S}) < \text{Var}(S_T)$$

Lower variance → lower optionality value.

(Kemna & Vorst, 1990)

Delta via Bump-and-Reprice

$$\Delta_0 = -0.50$$

Computed using central difference:

$$\Delta \approx \frac{V(S + \epsilon) - V(S - \epsilon)}{2\epsilon}$$

Unlike European/American:

- No closed-form delta.
- Requires re-simulation.
- Computationally expensive.

Metric	Asian ATM Put	American Put
Price	\$8.11	\$13.03
Initial Delta (Δ_0)	-0.50	-0.47
Payoff basis	Arithmetic avg $\bar{S}$ over path	Terminal price S_t
Early exercise	Not applicable	Optimal at ITM boundary
Vol sensitivity	Lower (averaging dampens)	Standard
Hedge type	Path-dependent Δ (requires MC reprice)	Standard binomial Δ

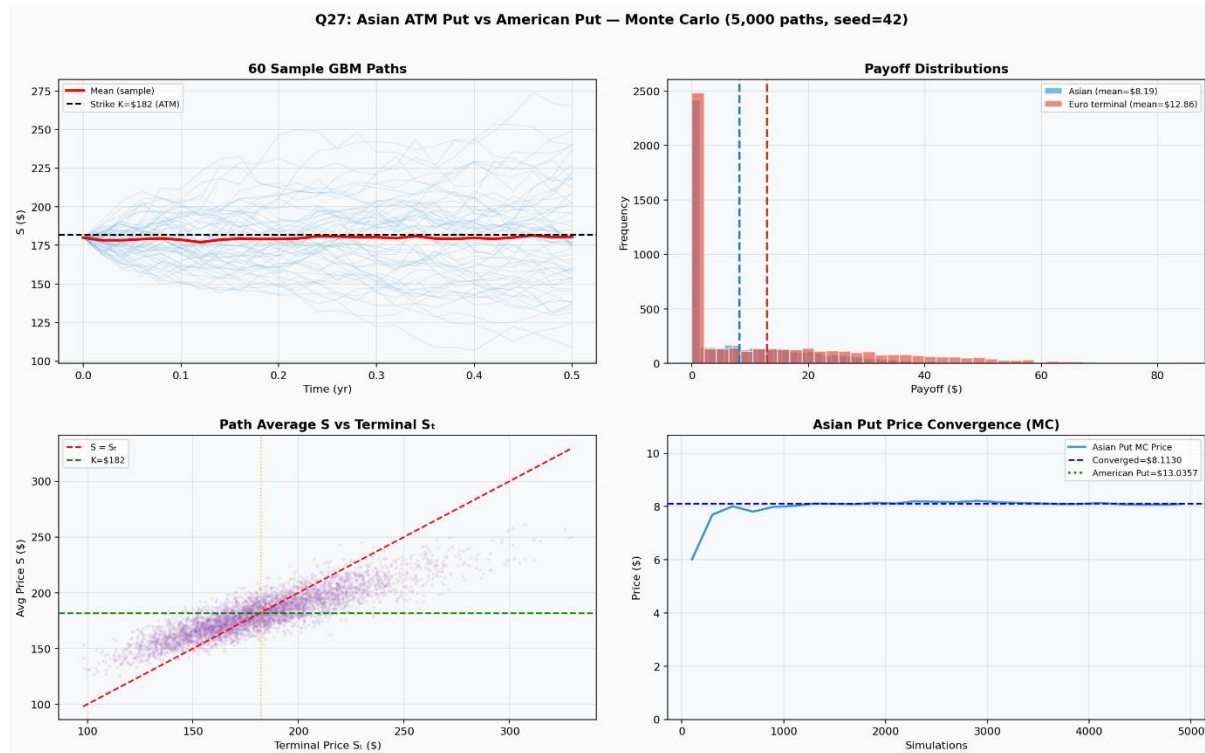


Figure 3: Top-left — 60 simulated stock paths. Top-right — payoff distributions (Asian averaging reduces right tail). Bottom-left — path average vs terminal scatter (averaging pulls extreme values toward mean). Bottom-right — Asian put MC price convergence vs American benchmark.

Commentary: Asian vs American Put

Feature	European Put	American Put	Asian ATM Put
Price	\$13.82 (3-step)	\$13.04 (25-step)	\$8.11 (MC, 5k paths)
Initial Δ	-0.4726	-0.4756	-0.5028
Hedge type	Standard Δ	Δ + early exercise monitor	Path-dependent Δ
Rebalancing	3 events	Up to 25 events	25 events, shrinking Δ
Delta stability	Moderate	Moderate	High — converges to 0 quickly
Final P&L (hedged)	≈\$0	≈\$0	≈\$0 (continuous reprice)

The Asian put is cheaper primarily because its payoff depends on the path average rather than the terminal price, and averaging reduces variance. The seller benefits from a lower initial premium collected but faces a more complex hedge: the delta is not available analytically and must be estimated numerically at each step via finite difference bump-and-reprice using Monte Carlo. As time passes, the running average becomes more "locked in" and the option's sensitivity to current price diminishes, causing delta to naturally shrink — which simplifies hedging in later periods but requires careful calibration early on.

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